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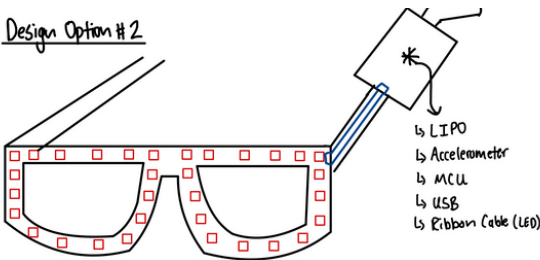
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MOTION CUES GLASSES

Problem Identification

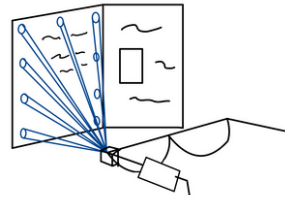
- Noticed car sickness while reading magazine on a road trip and explored hardware solutions to mitigate the sickness
- Drew inspiration from Apple's Vehicle Motion cues to create a similar wearable solution for books

Design Option #2



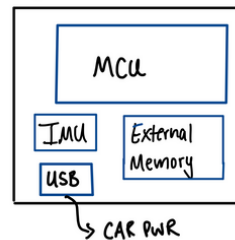
- LEDs in peripheral vision for proof of concept, more feasible
- Difficult to differentiate between car acceleration and head
- Bulky hardware on glasses temple

Design Option #1



- Projecting moving dots onto page required expensive hardware and complex systems like page size recognition and depth tracking
- Possibly iterate to this more complex design in future

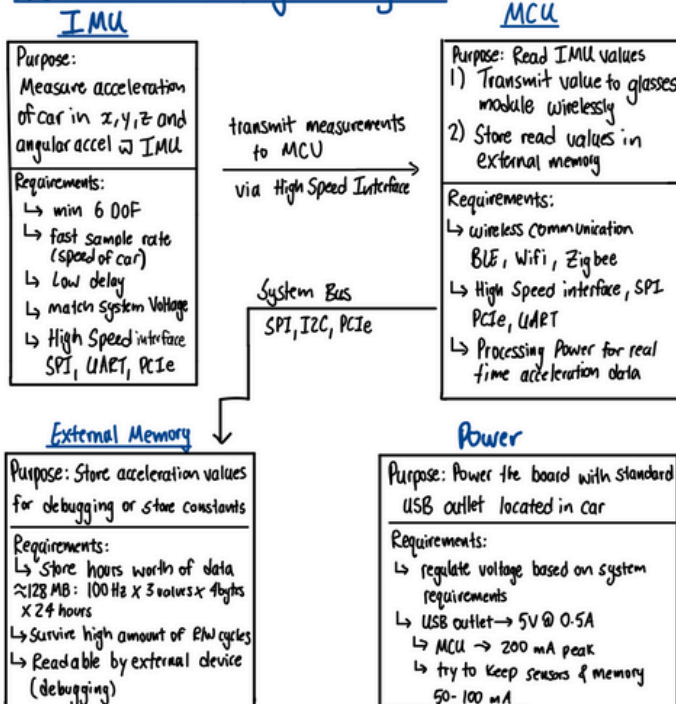
Design Option #3



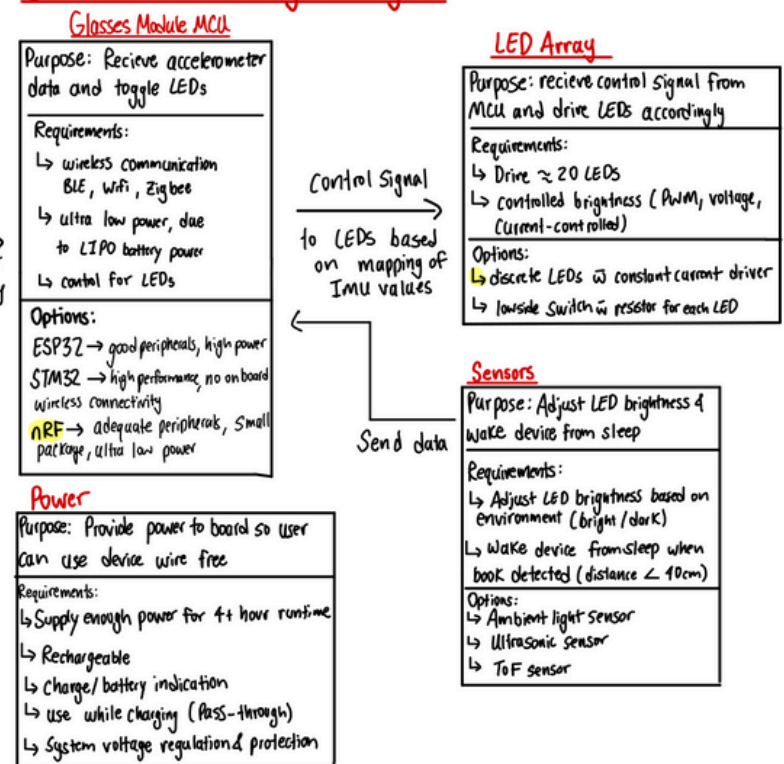
- Same design as option 2 but moved IMU to external board for more flexibility and reducing HW on temple
- Communication to glasses with BLE
- Opted for Flex PCB for glasses LED array

System Diagram

Accelerometer Board: System Diagram



Glasses Module Board: System Diagram



Goal: Sense Vehicle acceleration and provide corresponding visual cues to mitigate motion sickness in vehicle

- Vehicle mounted IMU module to capture linear and angular acceleration in 6-DOF interfaced over SPI
- Selected ESP32-C6 in IMU module to process and wirelessly transmit motion data over BLE
- Designed low-power glasses-mounted control board to receive motion data then translate to spatial LED cues in the user's peripheral vision
- Selected nRF52840 MCU for glasses module for its low power chip while maintaining wireless connectivity and a small footprint allowing for desired performance in the battery powered wearable

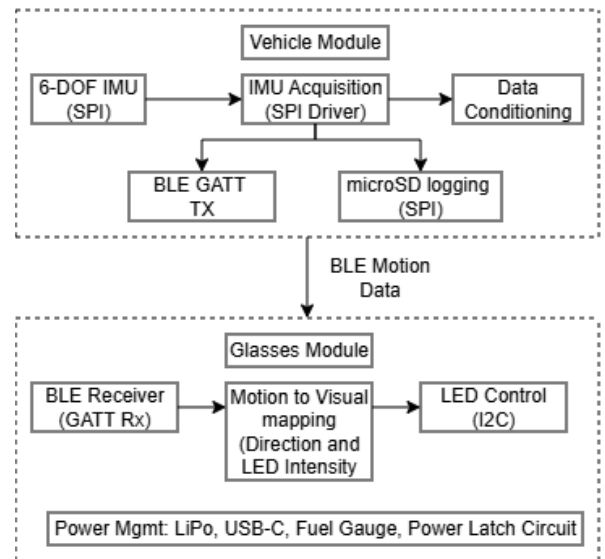
Project Features

- Two board system with with communication over **BLE**
- Vehicle module is 2-layer PCB using **ESP32-C6** with USB, SPI, BLE to capture and transmit IMU data
- Integrated Bosch BMI323 **6 axis IMU** with **16 bit** digital 3 axis accelerometer and 3 axis gyroscope
- Glasses module is to be designed into low power rigid-flex PCB assembly to receive data and map it to peripheral vision **LED array**
- Constant-current LED driver to control 16 LEDs with **PWM** over **I2C**
- LiPo battery power system on glasses module including **USB-C** charging, fuel gauging and soft-**power latch circuit**
- Removeable **microSD** storage via **SPI** to support logging and debugging of IMU values
- Interface protection, external crystal for deep sleep, and reset/boot controls for bringup, debugging and firmware development

Key Skills

- Embedded HW Design:**
 - System design across multiple MCUs (ESP32 & nRF)
 - BLE based inter-MCU communication
- PCB Design (Altium):** Multi-layer board stackups with RF and Power Integrity routing considerations
- Sensor Integration:**
 - IMU sensor which communicates over **SPI**
 - ALS and ToF sensor integration over **I2C**
 - Interrupt based alert signaling
- Protocols & Interfaces:** USB, SPI, I2C, BLE, microSD
- Power Electronics :**
 - LiPo Battery charging, monitoring and protection
 - Voltage regulator selection based on efficiency, dropout and quiescent current
 - Soft power latching circuit with PMOS and NMOS
- Debug and Bringup**
 - SWD programming and debugging
 - HW Validation during board bringup (future)

High Level Firmware Architecture (Vehicle and Glasses Nodes)



Glasses Module Board: 3V3 LDO Junction Temperature Calculation

3.3V LDO Junction Temp Calculation (Glasses Board)				
Device	Max Current Draw (mA)			
NRF52840 (BLE Active)	20			
VL53L4CD	24			
OPT4001	0.03			
LED Array (Adjustable)	80			
Total Current	124.03			
LDO: TLV70233DBVT				
I _{gnd} (mA)	0.37	P _D = [(V _{in} - V _{out}) × I _{out}] + (V _{in} × I _{ground})	P _D (W)	0.1505
V _{in} (V)	4.5		T _j (°C)	62.5048
V _{out} (V)	3.3			
T _a (°C)	25			
R _{ja} (°C/W)	249.2			

Glasses Module Board: I2C Pull-Up Resistor Calculation

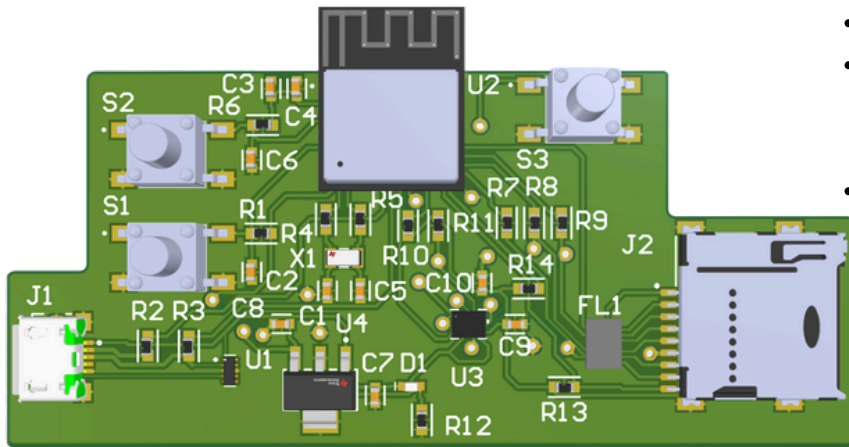
I2C Bus Capacitance Calculation					
Device	SCL (pF)	SDA (pF)	tr (ns)	tf (ns)	
VL53L4CD	4	10	300	300	
OPT4001	3	3	-	-	
PCA9635	10	10	1000	300	
MAX17048	60	60	300	300	
BQ25188YBGR	15	15	-	-	
Device Contribution C _{bus} (pF)	92	98			
Estimated Trace Contribution (pF)	100	100	<- Generous		
Estimated Total C _{bus} (pF)	200	200			
			V _{CC} (V)	3.3	
			V _{OL(max)} (V)	0.4	
			I _{OL} (mA)	3	
			R _{p(min)} (Ohms)	966.6667	
			tr (ns)	300	
			C _b (pF)	200	
			R _{p(max)} (Ohms)	1770.329	

$$R_p(\min) = \frac{V_{CC} - V_{OL(\max)}}{I_{OL}}$$
$$R_p(\max) = \frac{t_r}{(0.8473 \times C_b)}$$

Accelerometer PCB (Complete)

Design Goal: Vehicle mounted, USB powered PCB that is able to reliability transmit and store IMU data

- 2 Layer Rigid PCB stackup
 - L1 - Signal
 - L2 - GND
- 1 sided board (No major size constraint)
- To be mounted in enclosure and placed in area of minimal disturbance



Glasses Module PCB (In Progress)

Projected Design: Temple mounted Rigid PCB with ICs and major components, connected to Flex PCB with LED Array and detection sensors mounted to front of glasses

- 4 layer Rigid PCB stackup
 - L1 - Signal
 - L2 - GND
 - L3 - Power (VBAT/3V3/SYS)
 - L4 Signal
- Custom stripline Antenna for nRF52840 BLE
- 2 layer Flex PCB for LED Array & possible sensors
 - L1 - Signal
 - L2 - GND
- Connect between boards with FPC Ribbon cable

[ACCELERATION BOARD SCHEMATIC PDF](#)



[GLASSES MODULE SCHEMATIC PDF](#)



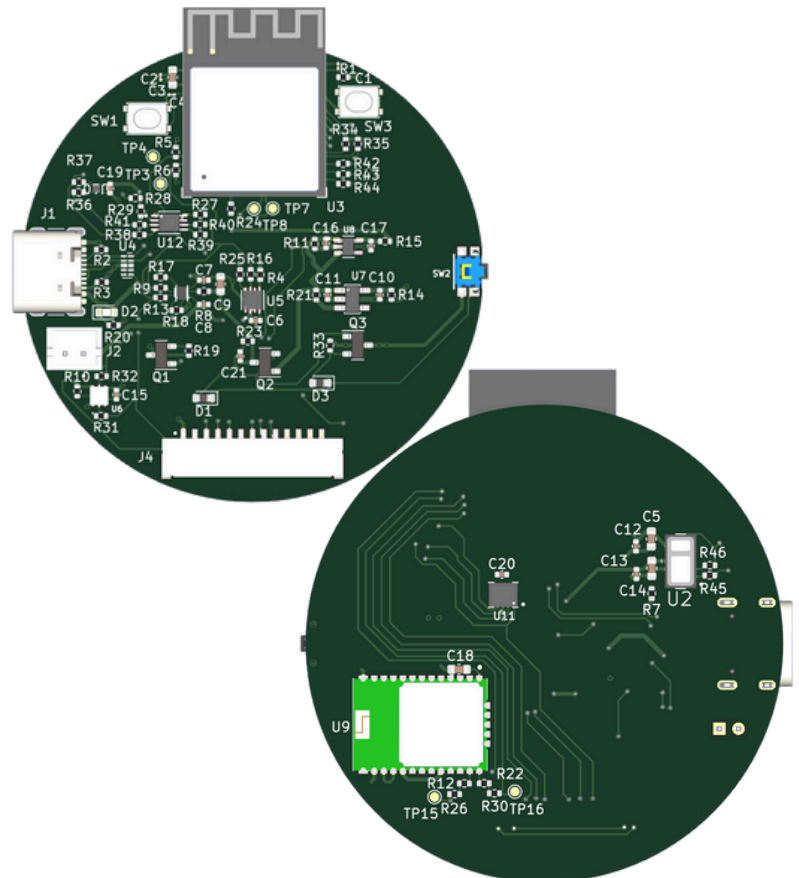
SMARTWATCH PCB PROJECT

Project Goal

- Design and develop a compact, low-power smartwatch integrating multiple sensors and wireless connectivity for health monitoring and activity tracking, with efficient power management and implemented with an 1.28" circular LCD

Key Skills

- **PCB Design (KiCAD):** 4-layer board layout in KiCAD with signal integrity, power distribution, and compact routing
- **Sensor Integration:** I²C and SPI communication with IMU, heart rate/SpO₂ sensor, GPS, compass, and ambient light sensors
- **Power Management:** USB-C charging, soft-latch power control, battery fuel gauging, and multiple voltage rails (1.8 V, 3.3 V)
- **Hardware Validation:** Bus topology design, level shifting, testpoint planning, and interface debugging
- **Protocols & Interfaces:** USB, I²C, SPI, UART, BLE, Wi-Fi, GNSS



- Designed a 1.28" circular 4-layer PCB integrating the **ESP32-S3-MINI-1** with Wi-Fi, BLE, and USB support
- Integrated sensors via I²C/SPI for heart rate (MAX30102), IMU (LSM6DSOX), magnetometer (AK0991x), GPS (MAX-M10S / CAM-M8Q), and ambient light monitoring, with appropriate **level shifting** for 1.8 V devices
- Implemented **USB-C** power input with battery charging, **power path** management, fuel gauge, and **soft-latch** power control circuit, LDOs for regulated 3.3 V and 1.8 V power rails
- Connected a 1.28" SPI TFT LCD with capacitive touch interface for user interaction
- Focused on signal integrity, proper **test point** and **jumper** placement, and **controlled impedance** routing for high-speed and RF lines

The diagram illustrates the system architecture for a smartwatch. The central component is the **MCU** (ESP32), which is connected to various peripherals and sensors.

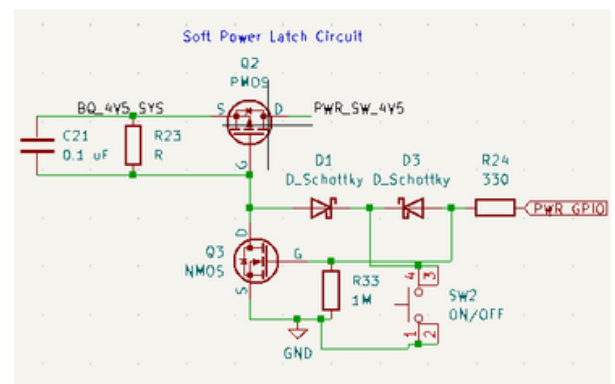
- Power Management:** The MCU is connected to a **Lipo Battery** and a **Battery Charge Circuit** (TP4056) via **USB-C** and **UART**. The Battery Charge Circuit is also connected to a **Bin1: Power** button.
- Display:** The MCU is connected to a **Display** (TFT OLED) via **I2C/SPI**. The display is also connected to a **Bin2: Scroll** button.
- Buttons:** The MCU is connected to a **Bin3: Select** button via **I2C**.
- Sensors and Modules:**
 - GPS:** The MCU is connected to a **GPS** module (uBlock Neo-GN, Quectel L78) via **UART/USB / I2C / SPI**.
 - Accelerometer:** The MCU is connected to an **Accelerometer** (MPU6050, ADXL345, LIS3DH) via **I2C**.
 - Heart Rate Sensor:** The MCU is connected to a **Heart Rate Sensor** (MAX30100, MAX30102) via **I2C**.
 - Other Modules:** The MCU includes **Nordic nRF52**, **STM32**, **Options**, **BLE?**, **Sleep Modes?**, **Debugging**, **to SWD, JTAG, UART**, and **I/O & Memory**.

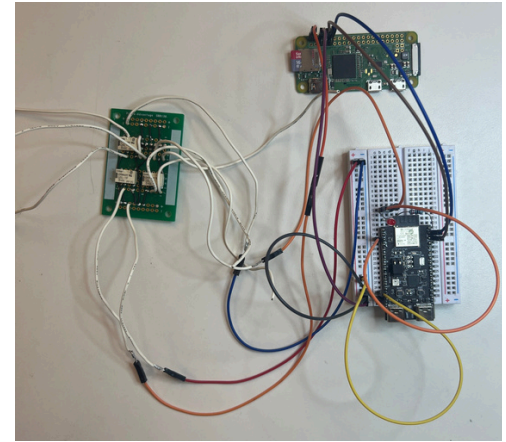
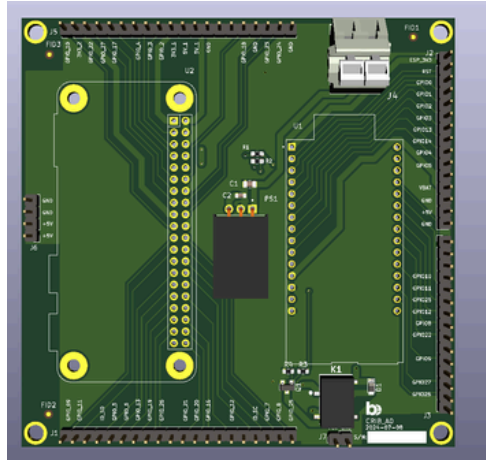
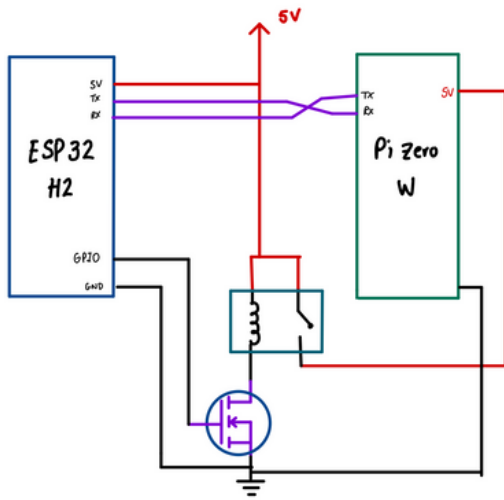
The diagram also includes a photograph of the physical smartwatch hardware, showing the display, battery, and various components on the PCB.

I2C Bus Capacitance Calculation						
Device	SCL (pF)	SDA (pf)	tr (ns)	tf (ns)	Vcc (V)	3.3
3.3V Devices					Vol(max) (V)	0.4
VEML6030	15	15	1000	1000	Iol (mA)	3
GC9A01	15	15	-	-		
MAX17048	60	60	1000	1000	Rp(min) (Ohms)	966.6667
BQ25188YBGR	15	15	-	-		
Device Contribution Cbus (pF)	105	105			tr (ns)	1000
Estimated Trace Contribution (pF)	20	20			Cb (pF)	125
Estimated Total Cbus (pF)	125	125				
					Rp(max) (Ohms)	9441.756
1.8V Devices						
AK09918	15	15	1000	300	Vcc (V)	1.8
MAX30102	15	15	300	300	Vol(max) (V)	0.36
Device Contribution Cbus (pF)	30	30			Iol (mA)	2
Estimated Trace Contribution (pF)	20	20				
Estimated Total Cbus (pF)	50	50			Rp(min) (Ohms)	720
					tr (ns)	300
					Cb (pF)	50
					Rp(max) (Ohms)	7081.317

		$T_J = T_A + (R_{\theta JA} \times P_D)$		
Max Load Current (A)	0.6			
LDO: AP2112K-3.3		$P_D = [(V_{in} - V_{out}) \times I_{out}] + (V_{in} \times I_{ground})$		
Ignd (A)	0.00008			
Vin (V)	4.5	Pd (W)	0.72036	
Vout (V)	3.3	Tj (°C)	94.1546	
Ta (°C)	25			
Rja (°C/W)	96			

Max Load Current (A)	0.15	$T_j = T_A + (R_{\theta JA} \times P_D)$ $P_D = [(V_{in} - V_{out}) \times I_{out}] + (V_{in} \times I_{ground})$	
LDO: MIC5365-1.8YC5-TR			
Ignd (A)	0.000039		
Vin (V)	4.5	Pd (W)	0.40518
Vout (V)	1.8	Tj (°C)	128.928
Ta (°C)	25	Note: will not be operating	
Rja (°C/W)	256.5	anywhere near max load current	





Project Goal

- Reduce the sleep current of the Pi Zero W from current 180 mA for battery powered applications
- Design for future expansion such as adding sensors, support for other MCUs
- Track time spent asleep during operation

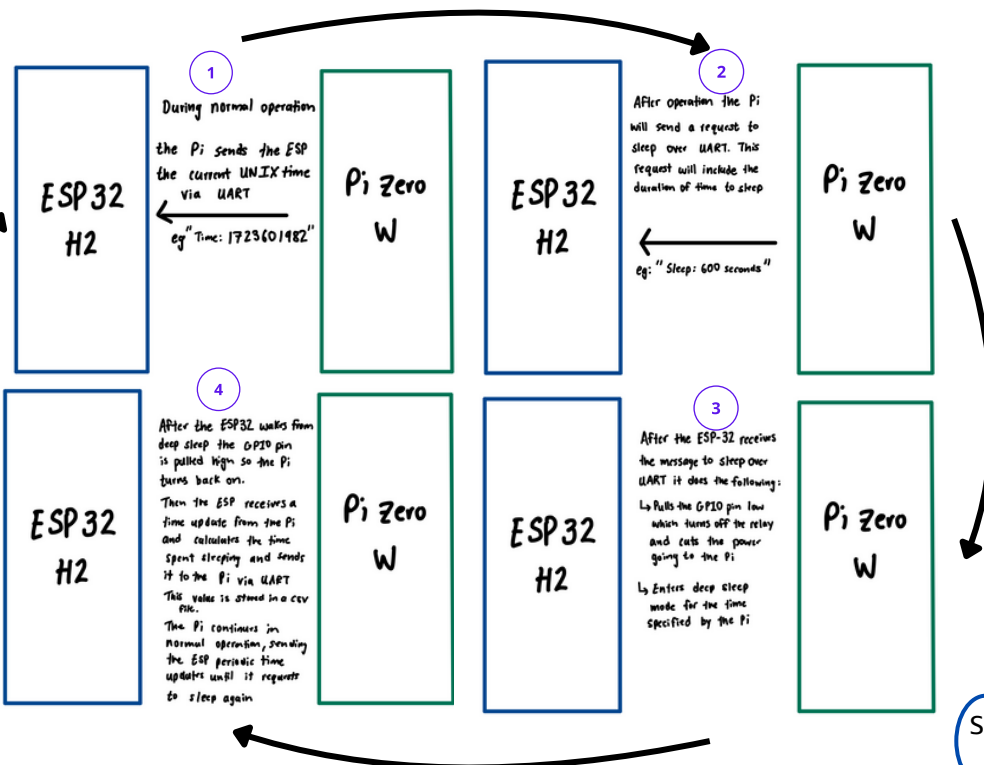
Key Skills

- PCB Design (KiCAD)
- ESP32
- UART (Software & Elec)
- C Programming
- FreeRTOS
- RTC
- Raspberry Pi
- Python

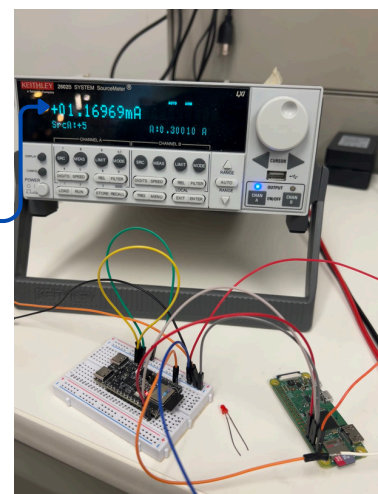
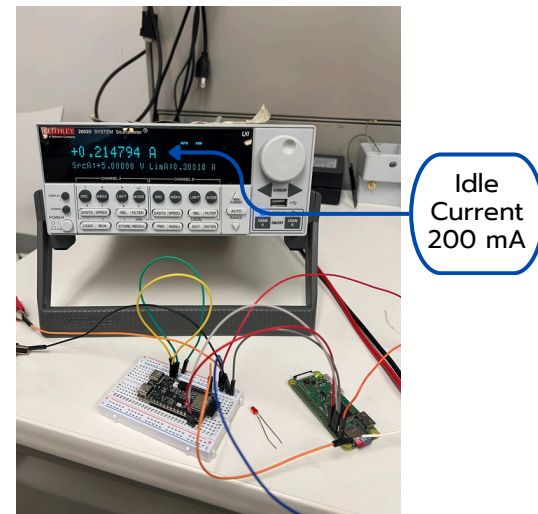
Project Features

- **UART** interface between ESP and Pi Zero
- Pi power controlled by Signal Relay
- Relay controlled by **MOSFET** and ESP GPIO
- On board 24V to 5V DC Converter
- Breakout Headers for expansion
- V2 PCB features support for Adafruit M0
- Sleep Current of **1 mA**

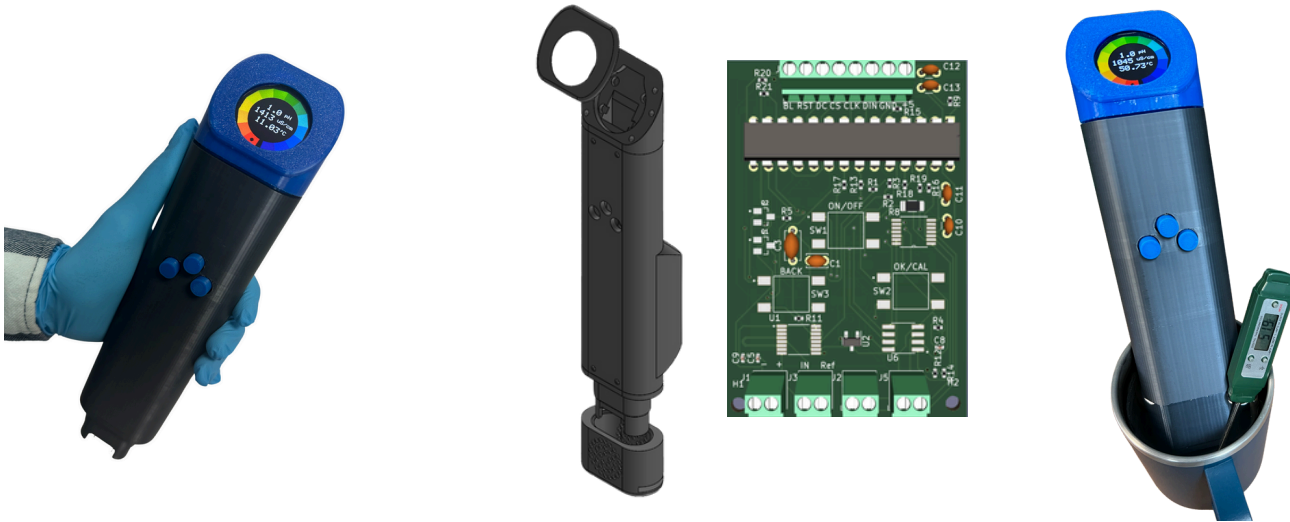
Sequence of Operation



Idle vs Deep Sleep Current



MULTI-PARAMETER WATER TESTER



Project Goal

- Design a handheld device capable of measuring the parameters of a water sample such as PH, Temperature, Conductivity and ORP
- Modular design such that it can be interchanged for different applications and parameters

Key Skills

- Low-Voltage PCB Design (KiCAD)
- High Resolution Sensor Measurement (ADC)
- Signal Amplification (Op-Amp)
- Digital Circuits (Latch Circuit)
- SPI (LCD)
- Fusion360
- C++
- ATmega328

Project Features

- 2 Layer PCB to interface sensors with MCU and LCD
- 12 Bit 4 channel **ADC** for increased resolution
- Power latch circuit for auto-shutoff
- Op-amp circuit for PH Sensor amplification
- Custom GUI to display measurements
- 3D printed enclosure for prototype

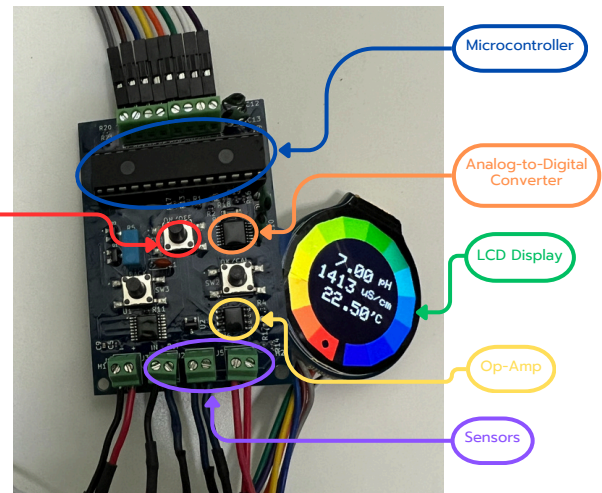
Version 1



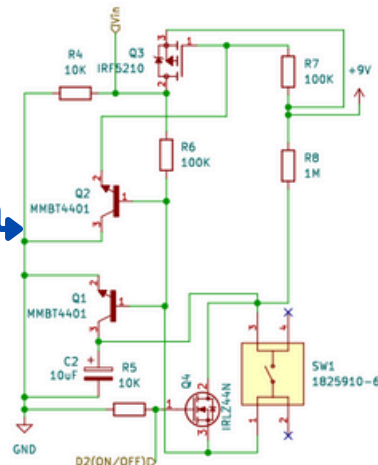
Increased Resolution with ADC

Shrunk PCB size with improved routing and SMT components

Version 2



Power Latch Circuit

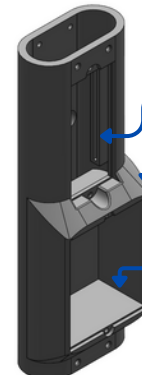


Designed Power Latching Circuit with power button that allowed for timed auto-shutoff of device, as a result saved battery usage

Version 1



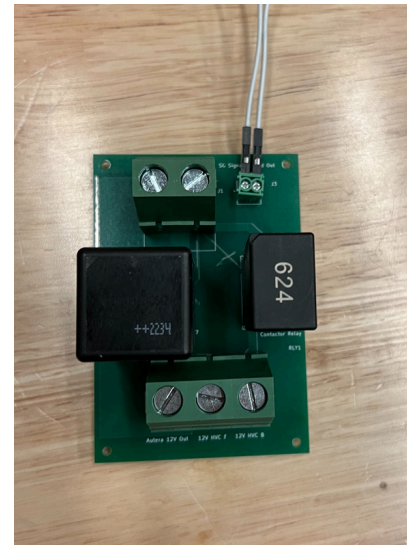
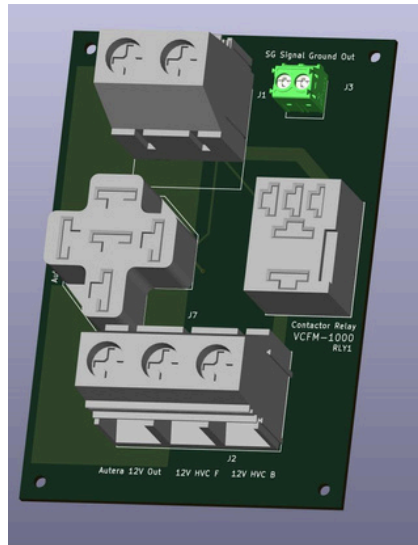
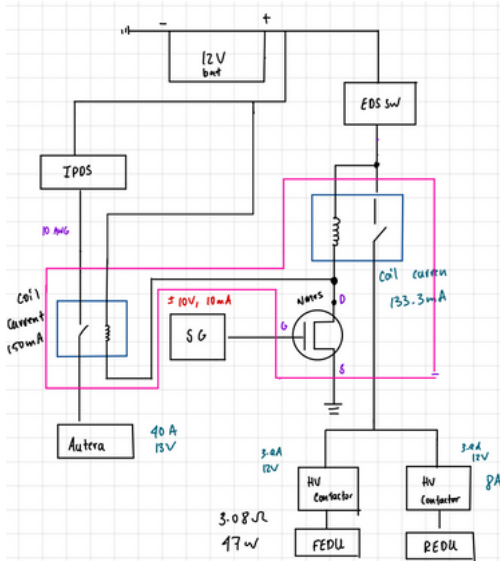
Version 2



Separate access to PCB for maintenance

Ergonomic and better fit in hand

Changed opening from 9V battery to 6xAAA for increased current capacity



Project Goal

- Design PCB capable of switching the power to the Autera and HV Contactors based on 3.3V signal from Speedgoat
- Be able to handle the max current draw of 40A and 8A respectively

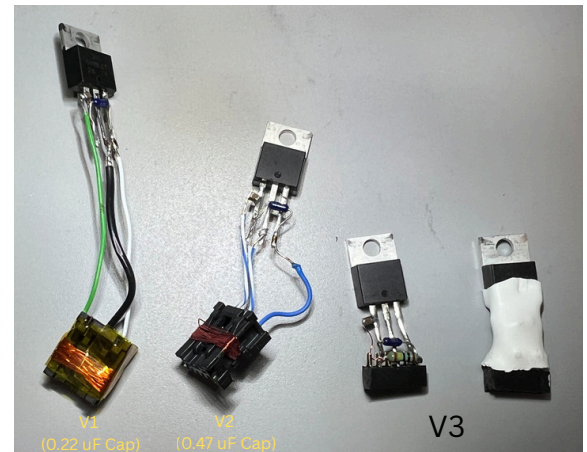
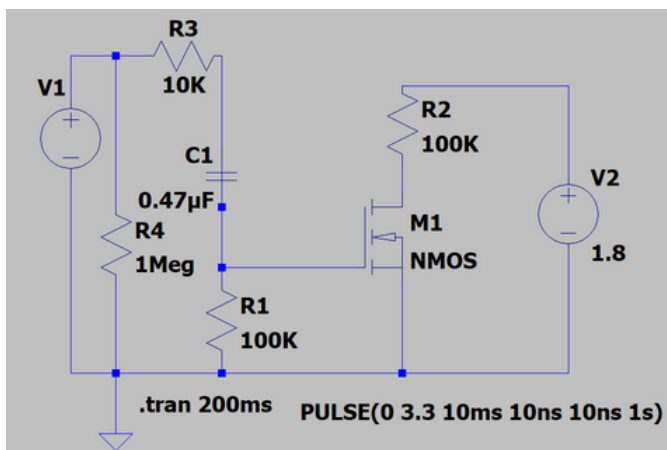
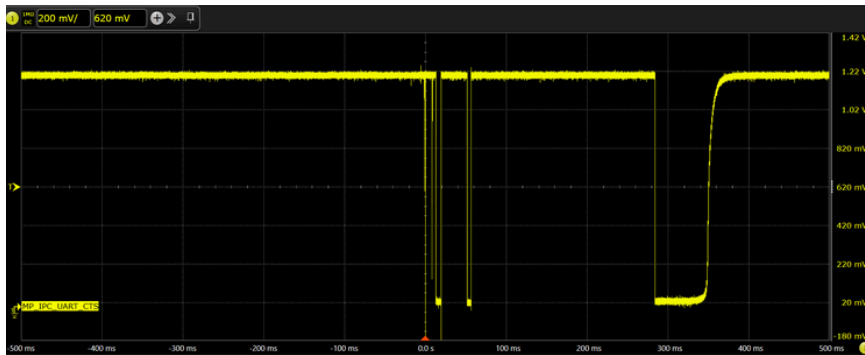
Key Skills

- High Current PCB Design (KiCAD)
- High Capacity Power Relay
- MOSFET Relay Control

Project Features

- Automotive grade Relays to control Power going to Autera and Contactors
- Large terminal blocks for 10 AWG wire
- nMOS to control relays from Speedgoat input

SIGNAL PULL DOWN DEVICE



Project Goal

- Design and create device capable of pulling 1.2V PS_HOLD signal low for > 50ms forcing device to RAM dump
- Eliminates the need SW team from having to manually short this signal on header with wire which is super unreliable and hard to time
- Minimize size and components (use existing, iterated from 2 FET design to 1

Key Skills

- SPICE Simulation
- Prototyping
- Soldering
- Signal Capture (Scope)

Project Features

- RC High Pass Filter to allow transient voltage for short controlled turning on of FET
- 1M bleed resistor to fix discharging issues
- Integrated into 2mm pitch header